



Hayes Diagnostics, Inc.

Investor Summary

Company

Hayes Diagnostics (HDI) is developing and commercializing next-generation diagnostic tests for hospitals, medical laboratories, and healthcare providers that need faster, more accurate pathogen identification than current methods can deliver. Founded by a globally recognized research scientist and a leading physician, HDI is built on advanced ultra high-resolution separation science for medical diagnostics. Its proprietary platform uses dielectrophoretic forces and a uniquely engineered microfluidic device to capture targets at locations directly linked to their identity. The company is initially focused on high-value clinical applications in sepsis and periprosthetic joint infection (PJI), where faster detection, resistance profiling, and severity assessment could meaningfully improve care. Following entry into these beachhead markets, HDI plans to expand its pathogen identification platform into the broader diagnostic market.

Problem

HDI addresses a critical gap in the \$24B+ global infection diagnostics market*, where legacy microbiology workflows remain too slow, labor-intensive, and often insufficiently reliable for timely clinical decision-making. These limitations delay targeted treatment, increase morbidity, raise the total cost of care, and contribute to antibiotic resistance. The current gold standard, microbial culture, can take up to two weeks to produce a result and may fail to identify the causative pathogen in up to one-quarter of implant-related infections. In high-acuity settings, these delays often force clinicians to treat empirically with incomplete information, which can lead to broader antibiotic use, avoidable side effects, longer hospital stays, and higher downstream costs.

Cause

Today's pathogen testing workflows depend on costly instrumentation, specialized personnel, and centralized laboratory infrastructure. Each handoff adds expense, complexity, and delay. While the market has accepted these tradeoffs for decades, that acceptance reflects a lack of viable alternatives rather than an optimal standard of care. HDI is positioned to challenge that status quo with a platform designed to improve both speed and accuracy while reducing operational burden.

Solution

HDI's patented platform is designed to rapidly and accurately identify bacterial pathogens based on their intrinsic electrical and physical properties. By reducing reliance on complex, multi-step laboratory workflows, the technology has the potential to simplify operations, lower labor requirements, and deliver actionable diagnostic results far faster than traditional culture-based methods. The result is a scalable diagnostic approach that supports earlier, more targeted treatment decisions, with the potential to improve patient outcomes while reducing overall healthcare costs.

Competition

Existing diagnostic solutions fall into three main categories:

- Microbial culture: highly trusted but slow, labor-intensive, and prone to failure.
- Molecular and immunoassay tests (PCR, ELISA): fast but expensive, infrastructure-heavy, and prone to false positives.
- Rapid strip or point-of-care tests: low cost and fast, but their long and costly development limit their suitability for multiple possible pathogens.

HDI's platform is positioned to deliver a compelling combination of speed, accuracy, ease of use, and development efficiency—creating the potential to outperform existing diagnostic options while reducing both test complexity and product development cost.

Customer Value Proposition

- Clinicians gain faster, more reliable diagnostic answers, enabling precise treatment decisions and reducing unguided antibiotic use.
- Hospitals and healthcare systems benefit from lower lab costs, reduced cost of care, and improved operational efficiency.
- Payers see reduced downstream costs from complications, prolonged care, and ineffective treatments.
- Patients experience faster diagnosis, better outcomes, and fewer adverse effects.

Business Development Progress

Hayes Diagnostics' device is currently at Technology Readiness Level (TRL) 4. The company has established an ongoing research partnership with a major medical device company focused on performance validation, and the platform is currently in use at a prominent West Coast university. Current work is generating foundational data to support future diagnostic test development and has reinforced market demand for faster, more accurate infection diagnostics. HDI is now advancing toward technology validation while preparing for strategic partnerships, pilot studies, and future regulatory milestones. Additional capital would accelerate development and compress time to market.

Financing Sought

The platform's current Technology Readiness Level reflects approximately \$9 million in grant-supported research and development. HDI is now seeking equity financing to advance the platform to TRL 9 and support commercialization. To learn more, see hayesdiagnostics.com or email info@hayesdiagnostics.com.

* Infectious Disease In Vitro Diagnostics Market (2025 - 2030), <https://www.grandviewresearch.com/industry-analysis/ivd-infectious-disease-market>